Levers and limits of active control in cross-situational word learning

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Abstract

Learners who seek the names of objects they cannot yet name can recover much of the cost that a Zipfian vocabulary imposes on learning from random input. Active machine learning distinguishes two sources of such benefit: allocating effort among separate problems, and choosing informative queries within one. We map a word learner's two levers onto these and relax earlier idealizations for two cross-situational learners, one eliminating candidate referents and one testing a single guess per word. (1) For the eliminative learner, choosing which word to hear yields a gain that grows only slowly with vocabulary size, whereas choosing what else is in view decides whether learning finishes at all once 100 referents are in view. (2) A learner active on a quarter of episodes gains much of the full benefit, and, as the sampling bias of active learning predicts, full control can entrench errors: fully active guess-test learners often stalled where learners with some passive input did not. Choice sets that over-represent frequent words can slow learning more than tenfold, and a learner that can pursue only words it has already heard gains far less and can fixate without passive input. (3) Becoming active late is at most slightly better than spreading the same control evenly; losing control early is much worse. (4) On real image captions, staying with the current scene beats seeking a new one once seeking costs more than an episode or two. What a learner's control acts on matters more than how much control it has.

Keywords: Word learning; Cross-situational learning; Active learning; Self-directed learning; Zipf's law; Development

Levers and limits of active control in cross-situational word learning

Introduction

Children hear words in a world that is far from uniform. Both the words caregivers say and the objects infants see follow highly skewed, Zipfian distributions (Clerkin, Hart, Rehg, Yu, & Smith, 2017; Lavi-Rotbain & Arnon, 2023), and under such distributions a learner who takes input as it comes must wait a long time for the rarest words. In a companion paper (Hidaka, Torii, & Kachergis, in preparation; hereafter Paper 1), we showed that this wait multiplies the number of learning episodes needed by roughly $\ln W$ for a vocabulary of W words, and that a learner who influences its input can recover much of the cost. A learner choosing among scenes can make rare words more likely, and an incremental learner who targets words it does not yet know learns 2–10 times faster than a passive one by the time it knows 50–90% of its vocabulary.

Machine learning suggests that the benefit could be far larger, and also that it could be far smaller. There, an active learner chooses which examples to have labeled, and can sometimes reach a given accuracy with exponentially fewer labels than a passive learner. To locate a threshold on a line, for example, a learner who chooses where to query can bisect the interval, and needs about $\log(1/\epsilon)$ labels rather than $1/\epsilon$ (Dasgupta, 2011; Settles, 2012). Such gains are not guaranteed. They arise when the examples that would distinguish the remaining hypotheses are rare under passive sampling and the active learner can find them; for some problems no strategy for choosing examples does much better than passive sampling (Dasgupta, 2005; Hanneke, 2007). And a learner that chooses its own data on the basis of its current beliefs can fail to correct them, a sampling bias that practical algorithms counter by continuing to sample at random some of the time (Beygelzimer, Dasgupta, & Langford, 2009; Dasgupta & Hsu, 2008).

A word learner's control can act on either side of the first distinction. Choosing which object to hear named–the control Paper 1 studied, and what developmental accounts of active learning usually mean–changes how a learner's episodes are divided among words, but not how much each episode reveals about the word it is spent on. Its

benefit is the time a passive learner spends waiting for the words it needs, a cost that grows only slowly, roughly logarithmically, with vocabulary size (19-fold at 1,000 words and 28-fold at 60,000 in Paper 1). Choosing what else is in view when a word is heard instead changes how informative each episode is. Under a Zipfian distribution, informative situations–those without a word's frequent competitors–become rarer the more referents are in view, and this second kind of control can then decide whether a learner learns at all. The distinction is sharpest for a learner that identifies a word's referent by ruling out competitors, as many analyses of cross-situational learning assume (Blythe, Smith, & Smith, 2010; Vogt, 2012); a learner that tests one guess per word is held up instead by its own wrong guesses, and for it the sampling bias of full control matters more than either lever. Developmental studies of active learning have mostly asked how much control children exercise; the machine-learning view suggests asking first what that control acts on.

Paper 1's analysis, like most formal treatments of active learning, also idealized the learner's control in four ways. The learner chose only which object to hear named, not what else was in view. It chose on every episode, from the whole vocabulary, among all the objects it could not yet name. Its control was the same from the first episode to the last. And obtaining a situation was free. None of these is true of children. Children are active learners (Foushee, Srinivasan, & Xu, 2023; Gureckis & Markant, 2012): they direct their gaze, hands, and requests toward what interests them (Partridge, McGovern, Yung, & Kidd, 2015; Yu, Smith, Shen, Pereira, & Smith, 2009), prefer stimuli of intermediate rather than minimal or maximal complexity (Kidd, Piantadosi, & Aslin, 2012), and adapt their information search to the environment from about age three (Ruggeri, 2022). But their control is partial. Much of what they hear is not directed by them at all. They choose among the objects at hand, not among every object they will ever name. Their capacity for control grows with motor development, which reshapes what infants see and do (Iverson, 2010; Kretch, Franchak, & Adolph, 2014; Walle & Campos, 2014). And, like adults, they do not always choose well (Bazhydai, Wong, Altmann, Jones, & Westermann, 2026; Tauber, Dunlosky, Rawson,

Wahlheim, & Jacoby, 2013).

This paper asks how the benefit of active control depends on these features. We organize the question around four aspects of control.

Which lever. Caregivers and children shape not only what is named but what else is in view when it is named: in play, they often arrange for a single object to dominate the child's view (Suanda, Barnhart, Smith, & Yu, 2018; Yu et al., 2009), and children exclude already-known objects as candidate referents for a new word (Grassmann, Schulze, & Tomasello, 2015). Study 1 asks how control over the target word and control over the accompanying distractors combine, and how this depends on the number of referents in view. In machine-learning terms, target selection chooses which of many separate problems to spend an episode on, and context selection chooses the query within a problem.

How much, and over what. Study 2 asks how much of the benefit survives when the learner is active only some of the time, when it chooses only among a small set of available words, and when it prefers words of intermediate familiarity, as the Goldilocks effect suggests infants do (Kidd et al., 2012). It also asks which unnamed objects a learner should pursue, and how much of the benefit remains if the learner can pursue only words it has already heard. These constraints correspond to the main settings of active machine learning (Settles, 2012): a learner that seeks the name of any object it cannot name can, like one making membership queries (Angluin, 1988), reach words it has never encountered; a learner choosing among the objects at hand is restricted to a pool; and a learner that pursues only words it has heard resembles one that decides, as each example arrives in a stream, whether to query it. Sampling bias predicts that full control need not be best.

When. Infants begin with little control over what they experience and gain it as they learn to reach, crawl, and walk. Because active selection is worth more late in learning, when only rare words remain, the timing of control could matter in its own right. Study 3 compares developmental trajectories from passive to active, and the reverse, with constant levels of control. Active machine learners, too, are often started

from randomly sampled examples, because a learner with poor initial beliefs chooses poorly (Settles, 2012).

At what cost. Gaining a new situation takes effort; asking more about the current one may take less. This is the patch-leaving problem of foraging theory (Charnov, 1976), which has been applied to search in memory and other cognitive domains (Hills, Todd, Lazer, Redish, & Couzin, 2015). It is also the machine-learning question of whether to obtain another label for an example already seen or a label for a new one (Sheng, Provost, & Ipeirotis, 2008), and of how labeling costs change the value of active selection (Settles, Craven, & Friedland, 2008). Study 4 asks, using real image captions, when a learner should stay with the current scene and when it should move on.

Across the four studies, the same theme emerges: active control helps by directing the learner's episodes toward its current bottleneck, and different levers act on different bottlenecks. For an eliminative learner, target selection acts on the wait for rare words, and its benefit is bounded by that wait; context selection acts on how informative each episode is, and with many referents in view its benefit has no comparable bound. For a guess-test learner, the main hazard is instead the sampling bias of full control. Levers that act on different bottlenecks need not combine, and constraints that change where the bottleneck lies can change which lever matters. Several of our results follow from principles established in machine learning, once a child's control is mapped onto the kinds of query studied there. Our aims are to make that mapping explicit for word learning, to measure the size of each effect under conditions closer to children's, and to identify where levers interact in ways that a single kind of query does not reveal. Figure 1 summarizes the four aspects and the main result for each. The Supplementary Material gives implementation details, the mechanism analyses we summarize, and robustness analyses with a learning criterion the guess-test learner can observe.

General method

All studies use the two incremental learners of Paper 1, an eliminative learner (after Siskind, 1996; Vogt, 2012) and a guess-test learner (after Trueswell, Medina,

Hafri, & Gleitman, 2013). There are W words, each paired with one of W referents. In each episode the learner hears one word, the *target*, in a situation containing its referent and $C - 1$ distractor referents. The eliminative learner keeps, for each word, the set of referents present every time that word has been heard, and learns the word once a single candidate remains. The guess-test learner keeps one guess per word, proposing a referent from the current situation that no other word claims, discarding the guess if the word is later heard without it, and learning the word once its guess is correct. The learner itself cannot observe that its guess is correct, yet in this model a word counts as learned, and an active learner stops pursuing it, as soon as it is. In Paper 1, a version that instead treats a word as known only once its guess has been confirmed twice, which the learner can observe, kept most of the active advantage at $C = 10$ (4.4-fold, against 4.9-fold). We report the original version throughout, for comparability with Paper 1, and summarize which of its results survive the observable criterion in Studies 2–4 (details in the Supplementary Material).

Word and referent frequencies are Zipfian, $p_r \propto r^{-a}$ over frequency rank r, with $a = 1$ unless stated ($a = 0$ is uniform). Two choices define each episode:

- *Target selection.* A *passive* learner's target is drawn by frequency: it hears whatever happens to be named. An *active* learner instead brings into view an object it cannot yet name–choosing to play with the car rather than the already-named ball–and hears that object named. Because each word names exactly one referent, this is equivalent to choosing among the words not yet known, but the learner does not need to have heard those words: it needs only to know which objects it cannot name. By default the active learner chooses among these objects in proportion to their frequency; Study 2 also examines a *uniform* policy that ignores frequency, and a *word-driven* learner that can pursue only words it has already heard, so that every word's first exposure must come from passive input. For brevity, we describe target selection below in terms of the words being targeted.

- *Context selection.* Under *random* context selection, the $C - 1$ distractors are

drawn by frequency. Under *familiar* context selection, each referent's frequency is divided by one plus the number of words whose candidate sets still contain it, so that distractors tend to be referents the learner has already ruled out for most words. This formalizes the use of known objects to disambiguate new words (Grassmann et al., 2015), and is consistent with evidence that discourse and attention, not only visual presence, determine which referents are candidates (Frank, Goodman, & Tenenbaum, 2009; Frank, Tenenbaum, & Fernald, 2013). The guess-test learner has no candidate sets, so for it familiar context instead weights each referent's frequency by one plus the number of consecutive times its word's current guess has been confirmed, favoring referents of words the learner has repeatedly heard.

What each lever can gain. Let word w need n_w exposures before it is learned. Target selection cannot change n_w, only how many episodes pass between a word's exposures. An active learner that never targets a known word needs about $\sum_w n_w$ episodes, whereas a passive learner must wait until its rarest words have been heard often enough. Under a Zipfian distribution with $a = 1$ the rarest word is heard on a fraction $1/(W H_W)$ of episodes, where $H_W \approx \ln W$ is the Wth harmonic number, against an average of $1/W$. Even if every word needed the same number of exposures, the passive learner would therefore take about H_W times as many episodes, and somewhat more, because it must wait for the last of many rare words (Paper 1). The benefit of target selection thus grows only slowly with vocabulary size. Context selection instead changes n_w. With random contexts, an exposure rules out a competing referent only if that referent happens to be absent, so a word whose most frequent competitor is absent from a context with probability q needs on the order of $1/q$ exposures. Under Zipfian distractors, q falls at least exponentially with the number of referents in view (Study 1), and a learner that controls the distractors removes this term. These are the two kinds of control distinguished in the Introduction.

Unless stated, vocabularies have $W = 1{,}000$ words, and we record the number of episodes until 50%, 90%, and 99% of the vocabulary is learned, with 30 replications per

condition. The reference implementations are written in R. Simulations of the eliminative learner at $C = 100$, at $W \geq 10{,}000$, of the target-selection variants of Study 2, and of the ramps of Study 3 used a faster reimplementation in C++, validated against the R implementation on every condition both could complete (Supplementary Section S1). Each study simulates its own reference conditions with independent random seeds, so the same condition's mean differs slightly between tables (for example, 88,220–89,857 episodes for passive learning at $C = 10$ with random context); comparisons and speedups are computed within one set of runs except where noted. Where learning times are skewed or comparisons close, we ran 200 replications per condition, and 95% confidence intervals are percentile bootstrap intervals over runs. Code and results are available in the project repository (`corpusXSL/analysis`).

Study 1: Which lever? Target selection vs. context selection

Paper 1 varies one lever: which word is the target of an episode. But a learner or caregiver also influences what else is in view when a word is heard. This study crosses target selection (passive or active) with context selection (random or familiar), and shows that at larger context sizes the second lever, not the first, decides whether elimination-based learning can finish at all.

We simulated the eliminative learner at $C = 10$, well within Vogt (2012)'s reported range of tractability ($C/W \leq 0.1$; his C/M), and at $C = 100$, at its boundary, for uniform and Zipfian vocabularies of $W = 1{,}000$ words, and recorded episodes to learn 99% of the vocabulary. Runs were capped at 5×10^8 episodes, and capped runs are reported as such.

Results: context size decides which lever matters

Table 1 and Figure 2 summarize the results at $a = 1$. At $C = 10$ ($C/W = 0.01$), active target selection does most of the work (19.1x), and familiar-context selection adds only a modest further benefit (2.0x alone, 28.6x combined), because elimination is already fairly effective when the distractor set is small.

At $C = 100$ ($C/W = 0.1$), random-context learning does not merely slow down under a Zipfian distribution: it fails. No passive or active run with random distractors reached even half the vocabulary within 5×10^8 episodes. The reason is the eliminative learner's requirement that every competing referent be absent from at least one of a word's contexts. With 99 frequency-weighted distractors per context, the most frequent referent ($p \approx 0.13$ at $a = 1$) is absent from any given context with probability only about 5×10^{-10} (computed exactly[1]), so every other word would need on the order of 2×10^9 exposures merely to rule out that one competitor. Active target selection cannot help, since it changes which word is the target but not which distractors accompany it. Familiar-context selection, which does change the distractors, restores learning (446,962 episodes), and once distractors are controlled, active target selection adds a further 28-fold advantage (15,819 episodes). At $C = 100$, controlling what else is in view is a precondition for learning at all. This is the situation in which active learning theory predicts its largest gains: the informative queries, here contexts without the most frequent competitor, are too rare for passive sampling to supply, and only the lever that chooses queries can supply them.

This failure is a matter of degree of skew, not an all-or-none property of $C = 100$. Sweeping the Zipf exponent at $C = 100$ (Figure 3, right) shows passive random-context learning slowing steeply but continuously, from 9,904 episodes at $a = 0$ to 40,456 at $a = 0.5$, 1.45 million at $a = 0.7$, 47 million at $a = 0.8$, and beyond our 10^8-episode cap from $a = 0.85$ on, tracking the exact probability that the most frequent referent is absent from a given context (0.17 at $a = 0.5$, 2.6×10^{-4} at $a = 0.8$, 1.5×10^{-6} at $a = 0.9$). Active target selection helps throughout this range (e.g., 12x at $a = 0.8$) but succumbs in turn (38 million episodes at $a = 0.85$, beyond the cap at $a = 0.9$). Only familiar-context selection keeps learning fast as skew grows (at $a = 0.9$: 106,039 passive

[1] By the exponential-clock representation of weighted sampling without replacement: each referent j receives an independent clock $E_j \sim \text{Exp}(p_j)$, the context's distractors are the $C - 1$ referents whose clocks ring first, and referent r is absent iff at least $C - 1$ other clocks ring before E_r. The resulting one-dimensional integral over a Poisson-binomial tail agrees with direct simulation wherever the probability is large enough to simulate; see `corpusXSL/analysis/c100_absence_probability.R`.

and 10,646 active episodes).

Context size, not the ratio C/W

The same failure appears at larger vocabularies (Table 2). $C = 100$ at $W = 10{,}000$ has the same ratio, $C/W = 0.01$, as $C = 10$ at $W = 1{,}000$, where learning is easy, and $C = 100$ at $W = 60{,}000$ has a smaller ratio still (0.0017). Yet at both sizes, with random distractors, no run, passive or active, learned even 10% of the vocabulary within 10^8 episodes (10 replications per condition). With 99 distractors per context, the most frequent referent ($p \approx 0.10$ at $W = 10{,}000$ and 0.086 at $W = 60{,}000$) is absent from a given context with probability only about 1.5×10^{-6} and 2.7×10^{-5} respectively, so the rarest words would need on the order of 10^{10} episodes or more merely to rule out that one competitor. Familiar-context selection again restores learning. Under Zipfian distractors, then, whether elimination is feasible depends on the absolute context size together with the skew of the referent distribution, rather than on the ratio C/W alone.

C/W does, however, shape how large the active advantage is once learning is possible. With familiar context at $C = 100$, the active advantage is 28-fold at $W = 1{,}000$ but only 7.6-fold at 10,000 and 9.3-fold at 60,000. Recording how many exposures each word needed before it was learned (10 replications per condition) shows why. Target selection does not change how many exposures a given word needs: passive and active learners need the same number, frequency band by frequency band. What differs is how each learner pays for them. An active learner never spends an episode on a word it already knows, so its total is essentially the sum of all words' needs, and under familiar context at $C = 100$ that sum is dominated by the most frequent words, whose competitors are other frequent referents: the top 1% of words need a median of 109–229 exposures each, the bottom half only 5–7. A passive learner instead waits for its last few rare words to be sampled often enough, so each exposure a rare word needs costs it roughly the inverse of that word's sampling probability. At $W = 1{,}000$ these last words are unusually hard–the last 1% learned need a median of 34 exposures, against 5–6 at $W = 10{,}000$ and 60,000–which inflates the passive learner's time, and

hence the speedup. Varying C at fixed W indicates that this hard tail grows with the share of all referents present in each context (5 replications per condition): at $W = 1{,}000$, the last 1% of words need a median of 3, 6, and 34 exposures at $C/W = 0.01$, 0.03, and 0.1, and at $W = 10{,}000$, 6, 10, and 297 at the same ratios. When each context contains a tenth of all referents, a competitor that survives one exposure has a good chance of reappearing in the next.

A learner that does not need full elimination

The guess-test learner of Paper 1, which keeps a single guess per word and counts a word as learned once its guess lands on the true referent, never needs every competing referent to be absent from some context. Accordingly it still learns at $C = 100$, $a = 1$ with random distractors (326,494 episodes passive, 176,578 active), where the eliminative learner never finishes, and active target and familiar-context selection give it only modest speedups (1.8x and 3.8x). Rare informative contexts do not hold this learner up, so neither lever has much to act on. What holds it up instead are its wrong guesses, which block the words whose referents they hold and, under full control, are not revisited: combining active target selection with familiar-context selection often prevented it from finishing (22 of 30 runs in Study 2, against none passive), and with a learning criterion the learner can observe, no fully active guess-test learner finished at $C = 100$. Study 2 examines this failure.

Cross-word mutual exclusivity

Every learner above narrows a word's candidates independently. Reisenauer, Smith, Smith, and Blythe (2013) show that a cross-word mutual exclusivity constraint–confirming one word's meaning immediately excludes that referent for every other still-unlearned word–can rescue an elimination-based learner from exactly the slowdown seen at large C. We added this rule to the eliminative learner, with cascades of further learning events allowed within an episode, and compared it against the independent model at $W = 1{,}000$, $a = 1$, random context (Table 3; 30 replications per

cell at $C = 10$, and 200 at $C = 100$, where learning time is heavy-tailed; no replication was censored).

At $C = 100$, mutual exclusivity delivers the predicted rescue: both passive and active learners now reach 99% of the vocabulary, in well under a million episodes on average. The trajectory is unusual, however. Essentially the whole vocabulary is learned in a single cascade: in 396 of 400 replications, the 50%, 90%, and 99% checkpoints were all reached within 10 episodes of one another, after a long initial period in which almost nothing was learned. And active target selection confers little benefit: 686,349 versus 771,125 episodes (medians 441,212 and 572,126; $z \approx 1.3$).

Both features have the same explanation. Nothing can cascade until at least one word has been learned by ordinary elimination, which requires ruling out every competitor for that word–least severe for the most frequent word, whose own referent is the competitor otherwise hardest to rule out. Once that first word is learned, its referent is removed from every other word's candidate set, which unblocks the next, and so on. The wait for the first learning event dominates the total time, and during that wait active target selection is inert: its advantage comes from never spending an episode on an already-known word, and while no word is yet known, that restriction does nothing. Holding the random seed fixed, active and passive target selection produce identical target sequences until the first word becomes known. Mutual exclusivity and active target selection, both beneficial elsewhere, therefore do not compound at large C–a second case, alongside the guess-test learner's failure under combined active target and familiar-context selection, of levers that fail to compose.

Study 2: How much autonomy, and over what?

In Paper 1 and in Study 1, an active learner is idealized in two ways: it exercises choice on every episode, and whenever it does, it can choose among every object it cannot yet name. Real learners exercise control only some of the time; they choose among whatever happens to be available (the toys in the room, not every object they will ever name); and they may not simply prefer the least-known item, since infants

preferentially attend to stimuli of intermediate familiarity (the "Goldilocks effect"; Kidd et al., 2012). Which unnamed objects a learner pursues may also depend more or less on how common they are. And a learner may be able to pursue a word only once it has heard it–asking what a half-heard word means, say–rather than seeking the name of an object it has seen but never heard named. This study relaxes each idealization in turn. These constraints correspond to the settings of active machine learning described in the Introduction–queries about any object, choice from a pool, and pursuit of items that have already arrived in a stream–and the sampling bias of active learning predicts that full control need not be best.

Setup

We extended the eliminative learner's target selection in three ways; later subsections also compare frequency-weighted with uniform choice among unnamed objects, and object-driven with word-driven pursuit. *Partial autonomy*: each episode is active with probability active_prob $\in \{0.25, 0.5, 0.75\}$ and passive otherwise, a constant blend of active and passive selection. *Goldilocks targeting*: among not-yet-known words, the learner prefers those it has encountered about twice, weighting each by $\exp(-(n-2)^2/8)$ times its frequency, where n is the number of times the word has been the target so far. *Bounded choice*: on each active episode the learner may choose only among a choice set of $k \in \{100, 50, 20, 5\}$ words, drawn without replacement in proportion to frequency (so that commonly-encountered things are more likely to be at hand), preferring an unknown word within the set and making a passive draw from the set if it contains none.

We ran each condition, together with pure passive and fully active selection, in four settings for a Zipfian vocabulary of $W = 1{,}000$ words: $C = 10$ with random and with familiar context selection ($a = 1$), and two settings at $C = 100$. Study 1 shows that at $C = 100$, $a = 1$, the eliminative learner never completes with randomly-composed contexts, even under fully active selection, so there we use the two $C = 100$ settings in which learning does complete: familiar context at $a = 1$, and

random context at $a = 0.75$, the steepest skew at which random-context learning finished in every run. All conditions were simulated with the C++ reimplementation of the eliminative learner described in the General method (200 replications per condition at $C = 10$, 30 at $C = 100$).

Results

Table 4 and Figure 4 report each condition's episodes to learn 99% of the vocabulary relative to fully active selection in the same setting.

Partial autonomy shows steeply diminishing returns. In every setting, a learner who is active on only a quarter of episodes already captures most of active selection's benefit: measured on a log scale, between two-thirds and four-fifths of the full passive-to-active speedup (e.g., at $C = 10$ with random context, 7.1x of a possible 19.0x). Each further increment of autonomy buys less. Study 3 asks whether, beyond its amount, the timing of autonomy matters.

Goldilocks targeting matters in only one setting, and not because of intermediate familiarity. Preferring words encountered about twice makes no difference in three of the four settings (0.996–1.04x). At $C = 100$ with familiar context it nearly halves learning time (0.56x), but a mechanism test (Supplementary Section S2) shows that this gain mostly disappears (0.93x) once familiarity is computed from encountered words only. Its benefit there comes from spreading early exposures across more of the vocabulary, which speeds up familiar-context selection, not from targeting words of intermediate familiarity as such.

The cost of bounded choice depends on the setting. In three settings, restricting choice costs little until the choice set is small, and costs more the smaller it is (1.03–1.09x with 100 words to choose from, rising to 2.8–4.1x with 5). At $C = 100$ with familiar context, the pattern is different: even a choice set of 100 makes learning 5.5 times slower, and the cost is not monotone in the size of the set (11.3x with 20 words, 10.4x with 5).

A diagnostic isolates the cause of this last pattern. Drawing the choice set

uniformly rather than in proportion to frequency, and otherwise leaving everything unchanged, removes the cost almost entirely at $C = 100$ with familiar context: 15,124 episodes with a uniform set of 100 words and 15,894 with 20, against 15,876 for unrestricted choice (30 replications each). The cost therefore comes not from restricting choice as such but from choice sets that over-represent frequent words: when the set is drawn by frequency and the target is then chosen from it by frequency again, a rare word's chance of being targeted scales roughly with the square of its frequency, and rare words are rarely targeted at all. Together with the Goldilocks result, this suggests that at $C = 100$ with familiar context, learning is fastest when exposures are spread across the whole vocabulary, including its rare words.

Why spreading exposures matters: unseen words dilute the familiarity signal. Familiar-context selection weights each candidate distractor by one over one plus the number of words that still hold it as a possible meaning. A word never yet targeted has ruled out nothing, so it adds the same amount to every referent's count, and while many words are unheard, familiar-context selection behaves almost like random selection. Policies that concentrate targets on frequent words keep words unheard for longer–634 after 10,000 episodes with a frequency-weighted choice set of 100 words, against 63 with unrestricted choice–and so delay the benefit of familiar context. Computing familiarity from heard words only removes most of the differences between policies (the choice set of 100 words goes from 5.7 to 1.3 times slower than unrestricted choice, and Goldilocks targeting from 0.57 to 0.93) and halves the learning time of unrestricted active selection (Supplementary Section S2). The effect is confined to $C = 100$ because only there does familiar context decide whether elimination succeeds; at $C = 10$ it adds a factor of 2 (Study 1).

Which unnamed object to pursue. Our default active learner chooses among the objects it cannot name in proportion to their frequency; a uniform policy ignores frequency. For the eliminative learner with random context the policy made no difference (4,657 and 4,670 episodes at $C = 10$; Supplementary Section S3): the number of exposures a word needs does not depend on which other words are known, so any

learner that targets only unknown words takes about the total number of exposures needed, whatever their order. With familiar context at $C = 100$, uniform targeting halved the learning time (8,181 vs. 15,978 episodes), roughly doubling the active advantage over passive learning from 30-fold to 58-fold, because it leaves fewer words unheard (about 6 after 5,000 episodes, against 239) and so dilutes the familiarity signal less. The active advantages we report for familiar context use frequency-weighted targeting and are therefore conservative.

The guess-test learner

We repeated the study with the guess-test learner, at $a = 1$ in all four settings, since this learner finishes at $C = 100$ even with random context (Study 1). Runs were capped at 5×10^6 episodes. Table 5 gives mean episodes to learn 99% of the vocabulary. Learning times at $C = 10$ are skewed, with a long tail of slow runs, so we ran 200 replications per cell there.

Partial control can beat full control. With random context, partial autonomy again shows diminishing returns, though less steeply than for the eliminative learner: being active on a quarter of episodes gains 60% (at $C = 10$; 95% CI 57–62%) and 42% (at $C = 100$; 22–56%) of the full speedup on a log scale, against two-thirds to four-fifths for the eliminative learner. With familiar context, partial autonomy is as good as or better than full autonomy. At $C = 10$, being active on a quarter of episodes gains 99% (96–102%) of the full speedup, and learners active on half or three-quarters of episodes need 19–22% fewer episodes than fully active ones (6,364 and 6,091, against 7,841; the confidence intervals of the means do not overlap). At $C = 100$, full active selection fails: 22 of 30 runs did not finish, whereas every learner active on 25–75% of episodes finished, faster than passive learning (68,658–80,128 episodes, against 85,818).

Instrumented runs (Supplementary Section S4) trace this failure to sampling bias compounded by a feedback loop. For this learner, a word can be learned only when its own referent is not held, as a wrong guess, by another word, and at $C = 100$ such blocking is the norm. A fully active learner never targets known words, so

familiar-context selection, which weights each referent by how often its word's guess has been confirmed, comes to reflect only the surviving wrong guesses. Contexts then contain exactly the referents those guesses claim, and the guesses confirm themselves. Spending a quarter of episodes rehearsing known words let every instrumented run finish, faster than passive or partially active learning.

Spreading targets helps. Goldilocks targeting more than doubles the speed of guess-test learning in three of the four settings (3,589 vs. 8,007 episodes at $C = 10$ with random context, 3,562 vs. 7,841 with familiar context, and 63,772 vs. 179,983 at $C = 100$ with random context), and uniform targeting does about as well at $C = 10$ and better at $C = 100$ with random context (43,144 episodes). Both fail, like frequency-weighted selection, at $C = 100$ with familiar context. The benefit comes from spreading targets over the unknown words, not from intermediate familiarity as such: frequency-weighted selection keeps returning to frequent words that are blocked and seldom targets the rarer words blocking them (at $C = 10$, 47–61% of its active episodes targeted a word already targeted ten or more times, against 2–5% for Goldilocks targeting). For this learner, frequency-weighted targeting–our default–is far from optimal, at least when the learner knows which of its guesses are correct.

Bounded choice is costly and erratic. At $C = 10$, frequency-weighted choice sets of 20–100 words give shorter median times than unrestricted choice (e.g., 5,753 vs. 7,610 episodes with 20 words, random context) but occasional very slow runs (with 50 words and familiar context, mean 27,861, median 6,232). At $C = 100$ with random context, sets of 20–100 words make learning 8.8–15.5 times slower than unrestricted choice, and a set of 5 only 2.8 times slower. This is the non-monotone pattern of the eliminative learner with familiar context, but with a different cause: drawing the set by frequency and then the target within it by frequency again makes targeting steeply favor frequent words, which for this learner are mostly blocked. Choosing uniformly within the same sets removes the cost (Supplementary Section S4).

Pursuing objects or pursuing words

Our active learner pursues objects it cannot name, and so can reach words it has never heard. A child could instead pursue a word: having heard an unfamiliar word, ask what it means, or watch for it to recur. Such pursuit is possible only after the first hearing. We therefore compared the object-driven learner with a *word-driven* learner, which on each active episode pursues a word it has heard at least once but not yet learned, chosen in proportion to its frequency, and otherwise behaves passively: when no heard word remains unlearned, its target is drawn by frequency. Every word's first exposure must therefore come from passive input. We ran both learners in the four settings above (eliminative learner: passive, object-driven, and word-driven in one set of runs, 200 replications at $C = 10$ and 30 at $C = 100$; guess-test learner: word-driven, 30 replications, compared with the passive and object-driven values of Table 5). As a benchmark we computed how many passive episodes it takes merely to hear 99% of the words once: 22,519 at $a = 1$ and 10,893 at $a = 0.75$ (means of 200 simulations).

At $C = 10$, most of the active advantage comes from reaching words never heard (Table 6). The word-driven eliminative learner is 3.4 times faster than passive learning with random context, against 19.2 times for the object-driven learner, and only 1.7 times faster with familiar context. Its learning time is close to the passive time to hear every word once plus the object-driven learner's time without first exposures: about $22{,}500 + (4{,}700 - 1{,}000) \approx 26{,}200$ episodes with random context, against 26,057 observed. Once a word has been heard, pursuing it removes the wait for its later exposures, but the wait for its first exposure remains and dominates. With familiar context the word-driven learner gains even less, because while many words remain unheard the familiarity signal is diluted (see above), and a word-driven learner keeps many words unheard for longest. The guess-test learner shows the same pattern: word-driven pursuit finishes just after the passive time to hear every word (24,035 and 23,456 episodes), 1.6 and 1.1 times faster than passive learning against 4.8 and 3.4 times for object-driven pursuit.

When the bottleneck is ruling out competitors, pursuing heard words is enough. At

$C = 100$, $a = 0.75$, with random context, the eliminative learner needs hundreds of exposures to rule out frequent competitors, and every word is heard long before it is learned. Word-driven pursuit there is only 3% slower than object-driven pursuit (10.6-fold vs. 10.9-fold over passive).

Without passive input, word-driven pursuit can fixate. In the remaining two $C = 100$ settings, fully word-driven learners usually did not finish, while passive and (for the eliminative learner with familiar context) object-driven learners did. None of the 30 eliminative runs with familiar context learned even 10% of the vocabulary in 5×10^8 episodes. Instrumented runs found the learner pursuing one of the first words it heard, whose frequent competitor almost never left the context, with no new word to turn to; the guess-test learner was stuck in the same way, on a word whose wrong guess was the referent of one of the most frequent words (Supplementary Section S5). A passive learner falls into the same traps for individual words, but moves on. Passive episodes remove the fixation by supplying new words to pursue. Eliminative learners pursuing heard words on 25%, 50%, and 75% of episodes, at $C = 100$ with familiar context, finished in 52,582, 73,267, and 133,950 episodes, against 431,500 passive; more pursuit was slower, the opposite of the object-driven learner (28,723, 21,191, and 18,047 episodes), because each word-driven episode is an episode not spent hearing a new word. Partially word-driven guess-test learners at $C = 100$ did as well as partially object-driven ones.

The contrast is the one between a machine learner that can query any instance (Angluin, 1988) and one that can query only instances that have arrived in its stream (Settles, 2012): only the first can reach parts of the input that passive sampling rarely visits. The two forms of pursuit thus suggest different roles for a child's own choices. Seeking the names of unnamed objects can shorten the wait for rare words; pursuing words already heard can only shorten the wait for repeat exposures, and depends on input the child did not choose to supply new words at all.

An observable learning criterion

The guess-test learner above stops pursuing a word as soon as its guess is correct, which it cannot observe. We therefore reran every guess-test condition with the lock-in criterion of Paper 1: a word is treated as known once its current guess has survived two consecutive exposures, and is scored as learned when that locked guess is also correct. Active selection pursues words whose guesses have not locked in, so a locked wrong guess can be corrected only on a passive episode. Supplementary Section S6 gives the full results (Table S3) and diagnostics.

Requiring confirmation strengthened the case against full control. At $C = 10$ the active advantage survived: by medians, fully active learning was 4.4 and 4.7 times faster than passive learning with random and familiar context (95% CIs 4.2–4.7 and 4.4–5.1), although 15 of 200 fully active runs with random context stalled short of 99%. At $C = 100$, no learner choosing among all unnamed objects on every episode finished, with either kind of context and whether it chose by frequency, by Goldilocks preference, uniformly, or among heard words only, while every passive and partially active learner did. Instrumented runs found chains of wrongly locked words, each holding the referent of another, which a fully active learner never revisits; with 100 referents in view, wrong guesses lock in by chance far more often (0.7–0.9% of wrong guesses at $C = 100$, against 0.03–0.13% at $C = 10$). Spreading targets still roughly halved median learning times at $C = 10$, but left many more runs stalled (with random context, 71 of 200 Goldilocks runs and 132 of 200 uniform runs, against 15 of 200 with frequency-weighted selection). Bounded choice became costlier at every set size, and fully word-driven learners failed even at $C = 10$.

Summary

Partial autonomy has diminishing returns for both learners, so much of active selection's benefit survives a learner who exercises it only occasionally. For the guess-test learner, and for a learner that can pursue only words already heard, full control can be worse than partial control, because a learner that chooses its data on the

basis of its beliefs does not collect the data that would correct them; passive episodes break this loop. When the bottleneck is hearing rare words at all, much of the benefit depends on being able to reach words never heard. The effects of Goldilocks targeting and bounded choice are more conditional: they depend on context size, on whether distractors are controlled, and, for bounded choice, on what the choice set contains rather than how large it is.

Study 3: A developmental passive-to-active trajectory

Studies 1 and 2 hold the degree of autonomy fixed. Infants' control instead grows: a newborn has little say over which scenes it experiences, and gains control as it learns to turn its head, point, reach, crawl, and walk. Active selection is worth more late in learning than early: for the eliminative learner at $C = 10$, the active advantage grows from about 4-fold at 50% of the vocabulary to 9-fold at 90% and 19-fold at 99% (Paper 1), because common words are learned quickly under either policy and only rare words remain late. A learner who becomes active over time might therefore do better than one with the same amount of control spread evenly, and much better than one who loses control over time. Active machine learners are, moreover, commonly started from randomly sampled examples, because a learner with poor initial beliefs chooses poorly (Settles, 2012); a passive start might likewise protect a learner from the errors of full control.

Setup

We made the probability that an episode is active a function of the episode count t, and compared a forward, linear ramp from passive to active, $\min(1, t/T)$, with its mirror image, a reverse ramp from active to passive, $\max(0, 1 - t/T)$. Over their first T episodes the two ramps are active equally often, but the reverse ramp places that activity where it is worth least. Both are indexed by episodes rather than by the fraction of the vocabulary learned, since motor development follows its own timetable rather than the child's vocabulary. The timescale T is therefore a free parameter, which

we varied from 1,000 to 20,000 episodes ($W = 1{,}000$, $C = 10$, $a = 1$, 200 replications per condition).

Results

Table 7 gives the reference points–pure passive and active selection, and constant blends at 13 intermediate levels of autonomy–and Figure 5 shows both ramp directions across seven timescales, $T \in \{1{,}000, 2{,}000, 3{,}000, 5{,}000, 8{,}000, 12{,}000, 20{,}000\}$ episodes.

Forward ramps are slower the longer the learner stays passive, rising smoothly from 4,869 episodes at $T = 1{,}000$, within 4% of pure active selection, to 10,612 at $T = 20{,}000$. Reverse ramps behave very differently. At $T = 1{,}000$ the reverse ramp takes 89,199 episodes, close to pure passive selection (89,852), and it stays near the passive level through $T = 8{,}000$ (57,301) before falling sharply by $T = 12{,}000$ (6,083). The threshold lies where the ramp's midpoint, $T/2$, is close to the time a fully active learner needs to finish (4,689 episodes). When T is large, a learner who starts active finishes before its autonomy has decayed far. When T is small, autonomy decays long before the vocabulary is complete, and the long remainder of the run is passive.

The contrast between directions at a given T confounds timing with the total amount of autonomy, because a learner who becomes active stays active for the rest of its run, while one who becomes passive stays passive. At $T = 1{,}000$, the forward learner is active on about 90% of its episodes and the reverse learner on fewer than 1%. To isolate timing, we compared each ramp with a constant blend of the same autonomy. For each run we computed its realized autonomy–the mean proportion of active episodes over the episodes it actually ran, which is exact because the ramp depends only on the episode count–and compared each ramp's mean episodes with the constant-blend curve of Table 7 at the ramp's mean realized autonomy, interpolating on a log scale. Confidence intervals for these ratios come from resampling runs within every ramp and blend condition. Forward ramps were 4–12% faster than constant blends of matched autonomy, and the gain grew with the timescale: 4% at $T = 1{,}000$ (autonomy 0.90), 11% at $T = 8{,}000$ (0.46), and 12% at $T = 20{,}000$ (0.26). Reverse ramps that lose

control before the vocabulary is learned ($T \leq 8{,}000$) were 1.2 to 2.3 times slower than matched blends, the largest cost at $T = 5{,}000$–$8{,}000$ (autonomy 0.03–0.07; ratios 2.23 and 2.28); those that finish while still mostly active ($T \geq 12{,}000$) were 2–4% slower. Every one of these 95% intervals excluded 1 and was within ±2% of its estimate. Timing therefore matters in the direction the checkpoint analysis predicts–autonomy is worth more late in learning than early–but for a learner who becomes active, its effect is modest compared with the effect of how much of the run is spent under control.

The guess-test learner

The guess-test learner shows the same pattern, less precisely. With the same ramps and a matching set of constant blends (200 replications each; passive 38,096 episodes, fully active 7,994), forward ramps were 1–14% faster than constant blends of matched autonomy, the gain again growing with the timescale: at $T \leq 2{,}000$, where the ramp is active on almost every episode, the difference was not reliable (ratios 0.99 and 0.96, 95% CIs 0.95–1.03 and 0.92–1.00), but from $T = 3{,}000$ on, forward ramps were 7–14% faster (ratios 0.93 to 0.86, every CI excluding 1). Reverse ramps were 1.2 to 1.8 times slower, most at $T = 8{,}000$ (1.84, CI 1.80–1.89). The threshold for reverse ramps lies at a longer timescale than for the eliminative learner, between $T = 12{,}000$ (20,868 episodes) and 20,000 (10,376), consistent with the $T/2$ rule given this learner's longer active completion time. Partial autonomy again shows diminishing returns: a constant blend with 25% active episodes captures 62% (59–64%) of the full speedup on a log scale.

With the observable learning criterion (Supplementary Section S7), losing control early was again costly (reverse ramps up to 2.3 times slower than matched blends), but becoming active late no longer reliably helped: forward ramps took 0.85–1.04 times as long as matched blends, with no trend in T. A passive start did protect against stalling, however. Fully active constant learners stalled in 20 of 200 runs, and forward ramps in 15 of 200 at $T = 1{,}000$ but in at most 1 of 200 at $T \geq 5{,}000$, although every ramp ends fully active.

In sum, a learner whose control arrives quickly relative to the time it needs to learn the vocabulary loses little of active selection's benefit, and is slightly more efficient (by up to 12–14%) than one with the same amount of control spread evenly, except for the guess-test learner with an observable criterion. Losing control early is much more costly.

Study 4: Staying or moving on: scene-switching cost on real image captions

Paper 1 tested target selection on Flickr8k (Hodosh, Young, & Hockenmaier, 2013): 8,108 photographs, each described by five captions written independently by different people, with a learner that always moved on to a new photo after each caption. Because each photo has several captions, these data also let a learner ask for another description of the scene in view rather than seek a new scene. This is the patch-leaving problem of the Introduction, and the machine-learning question of whether another label for an example already seen is worth more than a label for a new one (Sheng et al., 2008).

Setup

We treat each caption as a situation: a target word plus whichever other content words co-occur in that sentence, playing the role of distractors. After excluding function words (articles, prepositions, pronouns, and similar) via a hand-curated stoplist and words occurring only once, 4,318 content words remain, averaging 5.76 per caption–a naturally-occurring analogue of context size C, close to the $C = 10$ of our synthetic simulations. A word is eliminative-learnable if the intersection of the word-sets of every caption in which it appears narrows to the word itself; 83.3% (3,598 words) are learnable this way, and each of the remaining words keeps at most 8 candidates.

We compare three policies. *Passive*: sample a photo uniformly at random, reveal one of its captions at random, and always move on. *Active, move-on*: target the rarest not-yet-learned learnable word, sample a photo containing it, reveal one caption, and always move on. *Active, linger*: as move-on, but remain on the current photo–revealing another of its remaining captions–for as long as doing so is still shrinking some

unknown word's candidate set, moving on only once the current photo stops paying off. A `move_cost` parameter charges additional episodes each time an active learner moves to a new photo. It is the cost of *seeking* a scene that contains a particular word, formalizing the idea that eliciting another description of the scene already in view may be cheaper than finding a new scene built around that word. The passive learner does not seek scenes–it takes whichever photo comes next–so it is not charged; `move_cost` therefore compares the two active policies with each other, and measures how much the cost of search erodes their advantage over a learner that does not search. At zero cost, the two active policies differ only in what they observe.

Results

Table 8 and Figure 6 report mean episodes to reach 50%, 90%, and 99% of the learnable vocabulary (30 replications per cell). Paper 1 reports the passive vs. active move-on comparison at zero cost (23-fold at 99%); here we focus on the cost of switching. Even when each search costs ten extra episodes, the highest cost we tested, both active policies still finish 99% of the vocabulary in under half the episodes passive sampling requires.

Which active policy is better depends on the cost of switching. When switching is free or costs one episode, moving to a new photo wins (by 13% and 3% at 99%), because a new scene supplies more varied distractors than a second description of the same one. From a cost of two episodes on, lingering wins, and its advantage grows with the cost: 1% at a cost of 2, 3% at 3, 5% at 5, and 7% at 10.

The guess-test learner

We also ran the guess-test learner on the captions. Each caption updates every content word it contains, in random order: a word whose guess is absent from the caption discards it, and a word without a guess proposes a word in the caption that no other word claims. Learning was scored on the same 3,598 words. The eliminative learner's active policy, which always targets the rarest word not yet learned, deadlocks this learner: a rare word whose wrong guess appears in every caption containing it can

never be corrected, and a rarest-first learner keeps returning to it. The active guess-test learner therefore draws its target among unlearned words in proportion to their frequency, as the synthetic learners do, and keeps that target while it lingers on a photo. Runs were capped at 10^6 episodes (Table 9).

Three results differ from the eliminative learner. First, active selection does not complete the vocabulary: active runs level off at 97.5–98.9% of the learnable words, while every passive run reaches 99%. Captions are fixed, so a word that has guessed a word present in all of its captions is never corrected; passive sampling, which hears frequent words most often early on, presumably lets those words claim their own referents first and leaves rare words fewer wrong options. Second, at zero switching cost active selection helps much less than for the eliminative learner, 1.2, 2.3, and 2.7 times faster than passive learning at 50%, 90%, and 95% of the vocabulary, and once switching costs two or three episodes, active learning reaches 95% of the vocabulary later than passive learning does. Third, lingering matters much more. The policies are within 7% of each other at zero cost, but from `move_cost`=1 on, lingering reaches 95% of the vocabulary in 16–28% fewer episodes than moving on, and also ends with more words learned. The crossover thus lies between `move_cost` 0 and 1 for this learner, rather than between 1 and 2.

With the observable learning criterion (Supplementary Section S8), active runs leveled off lower (at 95.0–96.4% of the learnable words), while every passive run still reached 99%, and lingering was faster than moving on at every cost tested: at 90% of the vocabulary, by 3% when switching was free and by 22–37% at costs of 1 to 10 episodes.

General discussion

Paper 1 established that active control over learning situations can recover much of the cost of a Zipfian vocabulary. The four studies here ask what happens to that benefit when control is closer to what children have. The answers depend less on how much control a learner has than on which lever it acts on and where the learner's

bottleneck lies. Read against active learning in machine learning, the two levers produce the two kinds of gain that theory distinguishes, and the failures of full control are instances of the sampling bias it warns of.

Two kinds of control, two kinds of gain

An eliminative learner can be held up in two ways. It can wait for rare words to be heard often enough, or it can wait for frequent competitors to be absent from a word's contexts. Target selection removes the first bottleneck, since an active learner never spends an episode on a word it already knows. Context selection removes the second, since it fills contexts with referents already ruled out for most words. Which bottleneck dominates depends on the number of referents in view and on the skew of their distribution (Study 1). With 10 referents in view, competitors are easily ruled out, and target selection does most of the work. With 100, the most frequent referents are almost never absent from a context, and learning never finishes unless distractors are controlled–whether by familiar-context selection or by cross-word mutual exclusivity. Once distractors are controlled, target selection again adds a large benefit.

The two bottlenecks correspond to the two sources of benefit in active machine learning. Target selection reallocates episodes among words without changing what an episode reveals about its word, so it can save at most the time a passive learner spends on words it already knows, a saving that grows only slowly with vocabulary size. This is why its benefit, although large, is far from the exponential savings sometimes cited for active learning: it is closer to allocating a fixed sampling budget well than to choosing informative queries. Context selection is the lever that chooses queries. With 100 referents in view and Zipfian distractors, a context without a word's most frequent competitor is so rare that passive sampling almost never supplies one, which is the condition under which active learning theory predicts its largest gains (Dasgupta, 2005, 2011); controlling the distractors turned a learner that did not finish into one that learned the vocabulary in under half a million episodes (Study 1). For theories of children's active learning, this distinction suggests that the behaviors with the largest

effects may be those that shape what else is in view–a caregiver holding up a single object, or a child bringing one toy close so that it fills the view (Suanda et al., 2018; Yu et al., 2009)–rather than those that choose which object is named. The guess-test learner, which needs no competitor to be absent, gains little from either lever at $C = 100$ (1.8- and 3.8-fold; Study 1); its main hazard is the sampling bias discussed below.

This view also explains when levers fail to combine. Mutual exclusivity removes the competitor bottleneck by a cascade that begins with the first word learned; before that word is learned, target selection has nothing to act on, because no word is yet known, and it adds almost nothing (Study 1). Frequency-weighted choice sets restore the rare-word bottleneck that target selection removed, and they also delay familiar-context selection, because familiarity is computed over words, and words not yet heard dilute the signal (Study 2). For the guess-test learner at $C = 100$, active target selection and familiar-context selection, each helpful alone, together usually prevented learning from finishing, because the familiarity signal came to reflect the learner's own wrong guesses (Study 2). In each case, two forms of control that help separately interfere because one changes the state that the other depends on. Analyses of a single kind of query do not reveal such interactions, and they are the results here least anticipated by active learning theory.

Partial, constrained, and developing control

The most robust result is that partial control goes a long way. An eliminative learner active on a quarter of episodes gains between two-thirds and four-fifths of the full benefit, on a log scale, in every setting tested, and a guess-test learner 42–99% (Study 2). For the guess-test learner, partial control can also be better than full control, and with a learning criterion the learner can observe, no learner choosing among all unnamed objects on every episode finished at $C = 100$, while every partially active learner did. On real captions, too, a fully active guess-test learner stopped short of the whole vocabulary while a passive one did not (Study 4). This is the sampling bias of active machine learning. A learner that chooses its data on the basis of its current

beliefs never collects the data that would show those beliefs to be wrong (Dasgupta, 2011; Dasgupta & Hsu, 2008), and in empirical comparisons popular selection heuristics sometimes did worse than random sampling (Schein & Ungar, 2007). Algorithms with guarantees of consistency keep sampling every part of the input with some probability (Beygelzimer et al., 2009); passive episodes play that role here, by revisiting words the learner wrongly believes it knows. With familiar context there is a second loop, in which the learner's beliefs also shape which referents are in view, much as a model trained on labels it has itself predicted can entrench its own errors (Arazo, Ortego, Albert, O'Connor, & McGuinness, 2020); rehearsing known words on a quarter of episodes was enough to break it. Together with Paper 1's finding that a purely active learner cannot correct confident errors, this suggests that input outside the learner's own choices does more than dilute active control: it protects the learner from errors that full control entrenches. This is encouraging for the developmental relevance of active learning, because children's control over their input is intermittent: much of what they hear is overheard, initiated by others, or about things they did not choose.

Passive input matters most for a learner who can pursue only words it has already heard. Such a learner gains much less than one that seeks the names of unnamed objects when hearing each word at all is the bottleneck (Study 2: 3.4-fold rather than 19-fold at $C = 10$), and nearly as much when ruling out competitors is the bottleneck. Without passive episodes it can fixate on a word it cannot resolve and never hear another (for the guess-test learner with an observable criterion, even with 10 referents in view); with them, at $C = 100$ and familiar context, it did best when it pursued words least often. Whether children's active word learning is driven more by objects they cannot name or by words they have heard but do not understand is therefore not a detail: the first can bring rare words into the child's experience, while the second depends on others to do so.

What a child chooses among may matter more than how often they choose. Restricting choice to small sets of words costs little when the sets are drawn uniformly from the vocabulary, but a great deal when they over-represent frequent words, as the

objects within a child's reach plausibly do (Clerkin et al., 2017): an active learner is only as good as the pool it chooses from. Preferring words of intermediate familiarity (Kidd et al., 2012) depends on the learner. For the eliminative learner it made no difference in most settings, and where it helped, the gain came from spreading early exposures across the vocabulary. For the guess-test learner it more than doubled the speed of learning in three of four settings, because it spreads targets over the unknown words; spreading them uniformly did about as well with 10 referents in view and better with 100 (Study 2). With an observable learning criterion, both policies still learned most of the vocabulary about twice as fast, but left many more runs stalled by errors the learner never revisited (with random context, a third of Goldilocks runs and two-thirds of uniform runs). How a learner distributes its choices among the words it does not know can matter as much as whether it chooses them: uniform targeting also doubled the eliminative learner's active advantage with familiar context at $C = 100$ (Study 2), although it made no difference with random context.

Timing matters, in the direction expected from the growing value of active selection over learning, but modestly (Study 3). A learner who becomes active over time does up to 12–14% better than one with the same autonomy spread evenly, though not the guess-test learner with an observable learning criterion, and a learner who starts active and loses control before the vocabulary is learned does up to 1.8–2.3 times worse, depending on the learner and criterion. The largest effect of a developmental trajectory is through how much of the learner's run is spent under control, which depends on how quickly control arrives relative to how quickly the vocabulary is learned. Starting passive did help in one respect that resembles the practice of starting active machine learners from random examples: with an observable learning criterion, forward ramps that stayed passive for longer stalled less often than fully active learners.

The cost of new situations

On real captions, whether to stay with the current scene or seek a new one depends on the relative cost of the two (Study 4): when seeking a new scene is free or

cheap, a new scene is worth more than another description of the same one, and once it costs two or more episodes, staying is better. For the eliminative learner the difference was small, at most 7% at the costs tested; for the guess-test learner, lingering was 16–28% faster once switching cost anything, and with costs of two or three episodes per switch, active learning became slower than passive learning, which pays no cost of search. With an observable learning criterion, lingering was faster for the guess-test learner even when switching was free, and by 22–37% once it cost anything. The value of control therefore depends on what it costs to exercise, as it does for machine learners deciding whether another label for an example already seen is worth more than a label for a new one (Settles et al., 2008; Sheng et al., 2008). Like foraging animals deciding when to leave a patch (Charnov, 1976), a learner should stay with a scene while it still yields information at a higher rate than a new scene would, after the cost of finding one.

Limitations and future directions

Our two learners differ in instructive ways, but both are idealized, and our account of the two kinds of gain rests on them; other mechanisms, such as associative learners that accumulate graded co-occurrence strengths, may sit differently between the two. Study 3 uses a single vocabulary size, context size, and Zipf exponent, and Study 2 mostly a vocabulary of 1,000 words. Our mechanism tests for the guess-test learner, for word-driven fixation, and for stalls under the observable criterion rest on a few instrumented replications per condition. Our policies remain idealized: in most of our simulations the active learner knows exactly which words it has learned (for the guess-test learner, which of its guesses are correct; the analyses with an observable learning criterion relax this, with a fixed two-confirmation criterion), the ramp in Study 3 is linear and indexed by episodes rather than by age, and the costs in Study 4 are free parameters. We tested choice sets drawn by frequency or uniformly, but how real choice sets are composed, and how large they are, are empirical questions, and we make no claim about the timescale of infant development. Flickr8k captions are adult descriptions of photographs, not child-directed speech or a child's view. Each of these

idealizations could be informed by data. Egocentric recordings of infants' everyday experience (Clerkin et al., 2017; Sullivan, Mei, Perfors, Wojcik, & Frank, 2021) could be used to estimate how often children's attention is self-directed, which objects are available to choose among, how quickly self-directed exploration increases with motor development (Kretch et al., 2014), and how often a caregiver names a second object in the same scene rather than a new scene being sought. With such estimates, the present framework could predict not just whether active control helps, but how much each lever contributes over the course of a particular child's vocabulary growth.

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Table 1

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1000$-word vocabulary, eliminative learner, 30 replications per cell. Speedup is relative to Passive/Random at the same C. At $C = 100$, no random-context run reached even 50% of the vocabulary within the 5×10^8-episode cap, so speedups there are lower bounds.

	$C = 10$ ($C/W = 0.01$)		$C = 100$ ($C/W = 0.1$)	
Condition	Episodes	Speedup	Episodes	Speedup
Passive / Random	89,454	1x	$> 5 \times 10^8$	1x
Active / Random	4,695	19.1x	$> 5 \times 10^8$	–
Passive / Familiar	44,607	2.0x	446,962	>1,100x
Active / Familiar	3,127	28.6x	15,819	>31,000x

Table 2

Mean episodes to learn 99% of a W-word vocabulary at $C = 100$, eliminative learner. "$>$": no run reached even 10% (at $W = 1{,}000$, even 50%) of the vocabulary within the stated cap.

Condition	$W = 1{,}000$	$W = 10{,}000$	$W = 60{,}000$
Zipfian ($a = 1$), Passive / Random	$> 5 \times 10^8$	$> 10^8$	$> 10^8$
Zipfian ($a = 1$), Active / Random	$> 5 \times 10^8$	$> 10^8$	$> 10^8$
Zipfian ($a = 1$), Passive / Familiar	446,962	866,781	5,453,160
Zipfian ($a = 1$), Active / Familiar	15,819	113,584	589,504
Uniform, Passive / Random	9,904	80,048	424,968
Uniform, Active / Random	3,718	26,261	128,363

Table 3

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, eliminative learner, random context, with vs. without cross-word mutual exclusivity. Without mutual exclusivity, no $C = 100$ run reached even 50% of the vocabulary within 5×10^8 episodes, so those speedups are lower bounds.

Condition	Independent	Mutual exclusivity	Speedup
$C = 10$, Passive	89,454	23,184	3.9x
$C = 10$, Active	4,695	1,569	3.0x
$C = 100$, Passive	$> 5 \times 10^8$	771,125	>640x
$C = 100$, Active	$> 5 \times 10^8$	686,349	>720x

Table 4

Episodes to learn 99% of a $W = 1{,}000$-word Zipfian vocabulary, eliminative learner, relative to fully active selection in the same setting (first row: fully active episodes). Values above 1 are slower than fully active.

	$C = 10$, $a = 1$		$C = 100$, $a = 0.75$	$C = 100$, $a = 1$
Condition	Random	Familiar	Random	Familiar
Fully active (episodes)	4,692	3,132	628,691	16,060
Passive	19.0x	14.2x	10.6x	26.6x
25% active	2.67x	2.51x	2.33x	1.81x
50% active	1.67x	1.63x	1.58x	1.34x
75% active	1.24x	1.23x	1.21x	1.11x
Goldilocks	1.00x	1.04x	1.00x	0.56x
Choice set of 100	1.03x	1.09x	1.03x	5.51x
Choice set of 50	1.13x	1.15x	1.09x	8.01x
Choice set of 20	1.54x	1.43x	1.32x	11.3x
Choice set of 5	4.15x	3.32x	2.78x	10.4x

Table 5

Mean episodes to learn 99% of a $W = 1{,}000$-word Zipfian ($a = 1$) vocabulary, guess-test learner, 200 replications per cell at $C = 10$ and 30 at $C = 100$. ">" marks cells in which some runs did not finish within 5×10^6 episodes (number in parentheses); these runs are counted at the cap, so the mean is a lower bound.

	$C = 10$		$C = 100$	
Condition	Random	Familiar	Random	Familiar
Passive	38,322	26,748	328,648	85,818
25% active	15,044	7,963	255,740	68,658
50% active	11,531	6,364	237,110	71,732
75% active	9,166	6,091	200,699	80,128
Fully active	8,007	7,841	179,983	>3,718,842 (22)
Goldilocks	3,589	3,562	63,772	>4,835,958 (29)
Uniform	3,760	3,765	43,144	$> 5 \times 10^6$ (30)
Choice set of 100	11,180	13,776	1,589,313	>2,235,324 (7)
Choice set of 50	9,247	27,861	2,793,079	>3,582,840 (7)
Choice set of 20	6,670	9,331	>2,668,049 (4)	>3,570,502 (15)
Choice set of 5	11,095	8,785	501,200	350,426

Table 6

Mean episodes to learn 99% of a $W = 1{,}000$-word Zipfian vocabulary for passive, object-driven, and word-driven target selection, and the speedups of the two active learners over passive learning. Eliminative learner: one set of runs, 200 replications at $C = 10$, 30 at $C = 100$, cap 5×10^8 episodes. Guess-test learner: word-driven runs (30 replications, cap 5×10^6); passive and object-driven values from Table 5. ">": some runs did not finish (number in parentheses), counted at the cap.

		Active (episodes)		Speedup over passive	
Setting	Passive	Object-driven	Word-driven	Object	Word
Eliminative learner					
$C = 10$, $a = 1$, random	89,720	4,683	26,057	19.2x	3.4x
$C = 10$, $a = 1$, familiar	44,641	3,130	26,177	14.3x	1.7x
$C = 100$, $a = 0.75$, random	6,759,860	622,007	639,077	10.9x	10.6x
$C = 100$, $a = 1$, familiar	431,500	15,991	$> 5 \times 10^8$ (30)	27.0x	–
Guess-test learner					
$C = 10$, $a = 1$, random	38,322	8,007	24,035	4.8x	1.6x
$C = 10$, $a = 1$, familiar	26,748	7,841	23,456	3.4x	1.1x
$C = 100$, $a = 1$, random	328,648	179,983	>4,109,440 (23)	1.8x	–
$C = 100$, $a = 1$, familiar	85,818	>3,718,842 (22)	>3,802,207 (20)	–	–

Table 7

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, eliminative learner, $C = 10$, for constant levels of autonomy (proportion of episodes on which the target is actively selected); 200 replications each. Study 2 reports some of the same blends, relative to fully active selection, from a separate set of simulations.

Proportion active	Episodes	Proportion active	Episodes
0 (passive)	89,852	0.30	11,123
0.02	41,997	0.40	9,192
0.05	29,425	0.50	7,857
0.075	24,474	0.60	6,900
0.10	21,112	0.75	5,839
0.15	16,922	0.90	5,077
0.20	14,325	1 (active)	4,689
0.25	12,489		

Table 8

Mean episodes to reach 50/90/99% of the 3,598-word learnable vocabulary, Flickr8k captions, eliminative mechanism. `move_cost` *is the number of extra episodes charged each time an active learner seeks a new photo (the passive learner does not seek photos and is not charged).*

Condition	move_cost	50%	90%	99%
Passive	–	10,386	60,719	151,671
Active, move-on	0	2,779	5,643	6,528
Active, linger	0	3,125	6,511	7,515
Active, move-on	1	5,558	11,285	13,057
Active, linger	1	5,483	11,587	13,461
Active, move-on	2	8,338	16,928	19,585
Active, linger	2	7,841	16,664	19,408
Active, move-on	3	11,117	22,570	26,113
Active, linger	3	10,199	21,740	25,354
Active, move-on	5	16,675	33,856	39,170
Active, linger	5	14,915	31,893	37,247
Active, move-on	10	30,571	62,069	71,811
Active, linger	10	26,705	57,275	66,980

Table 9

Guess-test learner on Flickr8k captions: mean episodes to learn 50%, 90%, and 95% of the 3,598-word learnable vocabulary, and the proportion learned at the end of the run (cap 10^6 *episodes), 30 replications per cell. Passive learning reached 99% in every run (mean 165,506 episodes); no active run did.*

Condition	move_cost	50%	90%	95%	Final
Passive	–	7,746	39,349	60,015	99.3%
Active, move-on	0	6,291	17,165	21,937	97.9%
Active, linger	0	6,693	17,976	21,862	98.5%
Active, move-on	1	12,583	34,330	43,875	97.9%
Active, linger	1	9,631	29,663	37,030	98.5%
Active, move-on	2	18,874	51,496	65,812	97.9%
Active, linger	2	12,569	41,351	52,199	98.5%
Active, move-on	3	25,166	68,661	87,749	97.9%
Active, linger	3	15,507	53,039	67,367	98.5%
Active, move-on	5	37,749	102,991	131,623	97.9%
Active, linger	5	21,383	76,414	97,704	98.5%
Active, move-on	10	69,206	188,817	241,310	97.9%
Active, linger	10	36,073	134,853	173,547	98.5%

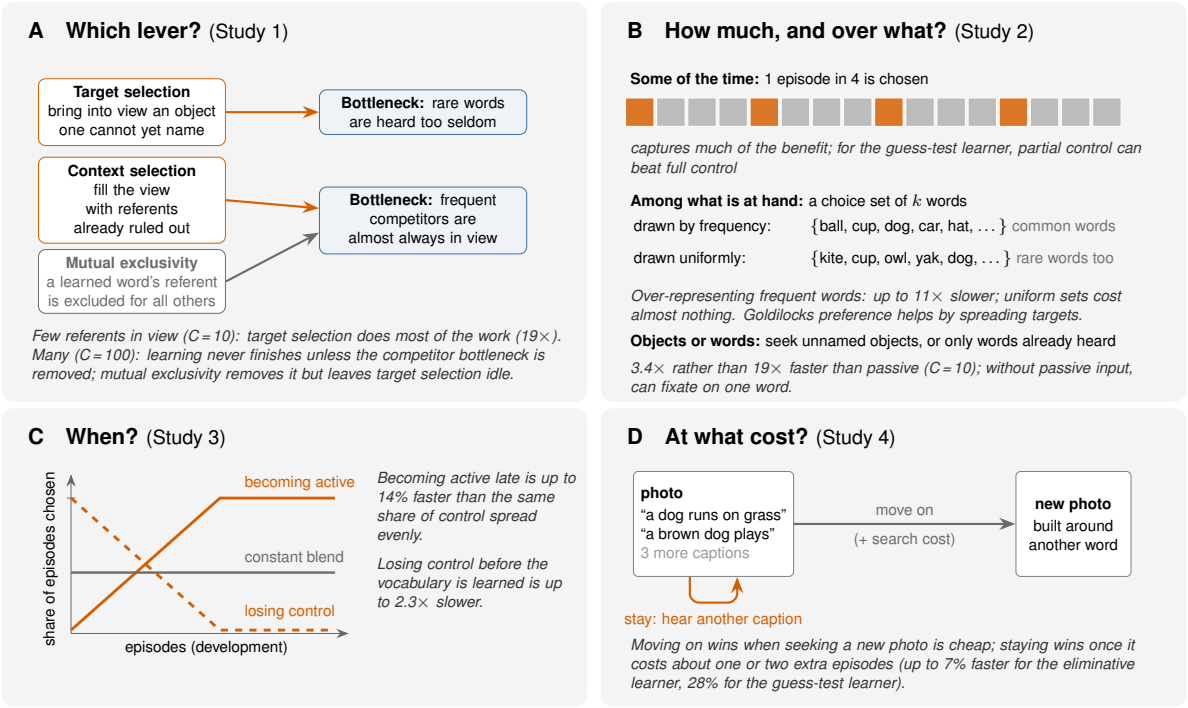


Figure 1. Overview of the four studies. (A) An eliminative learner can be held up by rare words that are heard too seldom, or by frequent competitors that are almost always in view; target selection acts on the first, and context selection or cross-word mutual exclusivity on the second. (B) A learner may exercise control only on some episodes, only over a set of words at hand, and only over words it has already heard rather than any object it cannot name. (C) Control may arrive or fade over development. (D) On real captions, a learner can stay with the current photo for another caption or seek a new one at a cost. Orange marks active choices, grey passive input or properties of the learner; italic notes give the main result for each study.

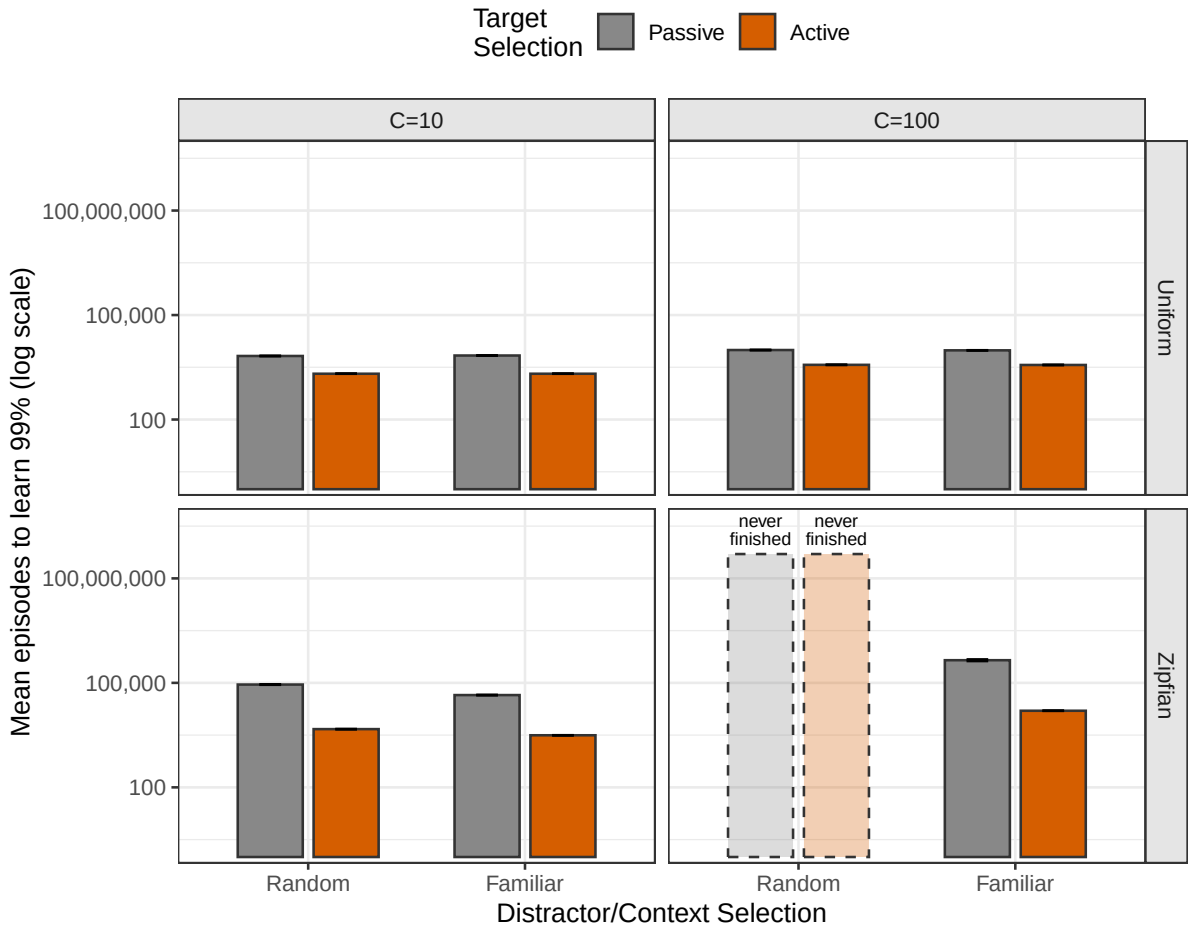


Figure 2. Mean episodes (log scale, ± 1 SE) for the eliminative learner to acquire 99% of a 1000-word vocabulary, by target selection (bar fill) and context selection (x-axis), for uniform and Zipfian ($a = 1$) distributions and $C = 10$ and $C = 100$. Faded, dashed bars mark conditions in which no run reached 99% (or even 50%) of the vocabulary within the 5×10^8-episode cap; they are drawn at the cap and are lower bounds.

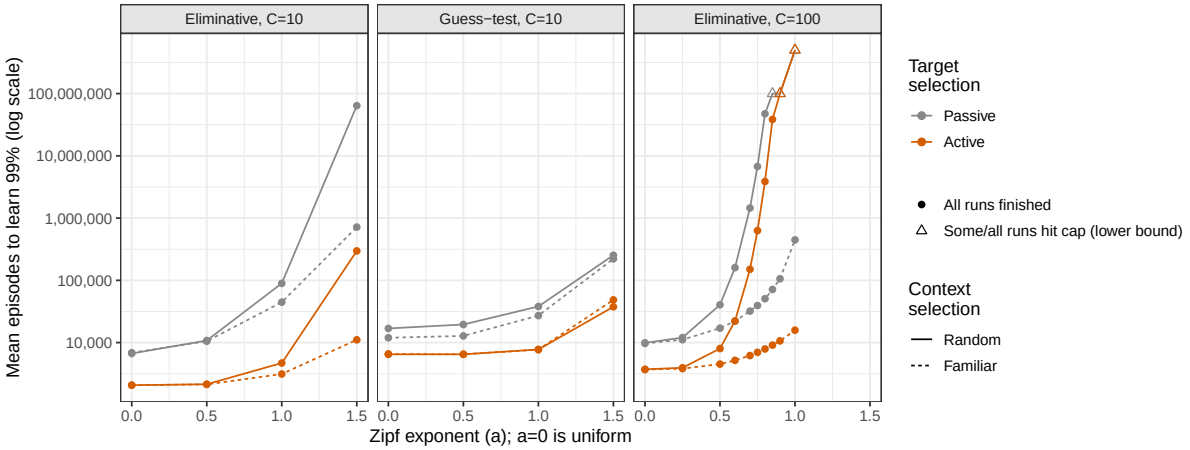


Figure 3. Mean episodes (log scale) to acquire 99% of a 1000-word vocabulary as a function of the Zipf exponent a, for the eliminative and guess-test learners at $C = 10$ and the eliminative learner at $C = 100$, by target selection (color) and context selection (line type). Open triangles mark cells in which some or all runs reached the episode cap (5×10^7 for $a \leq 0.75$, 10^8 for $0.8 \leq a \leq 0.9$, 5×10^8 for $a = 1$); these points are lower bounds.

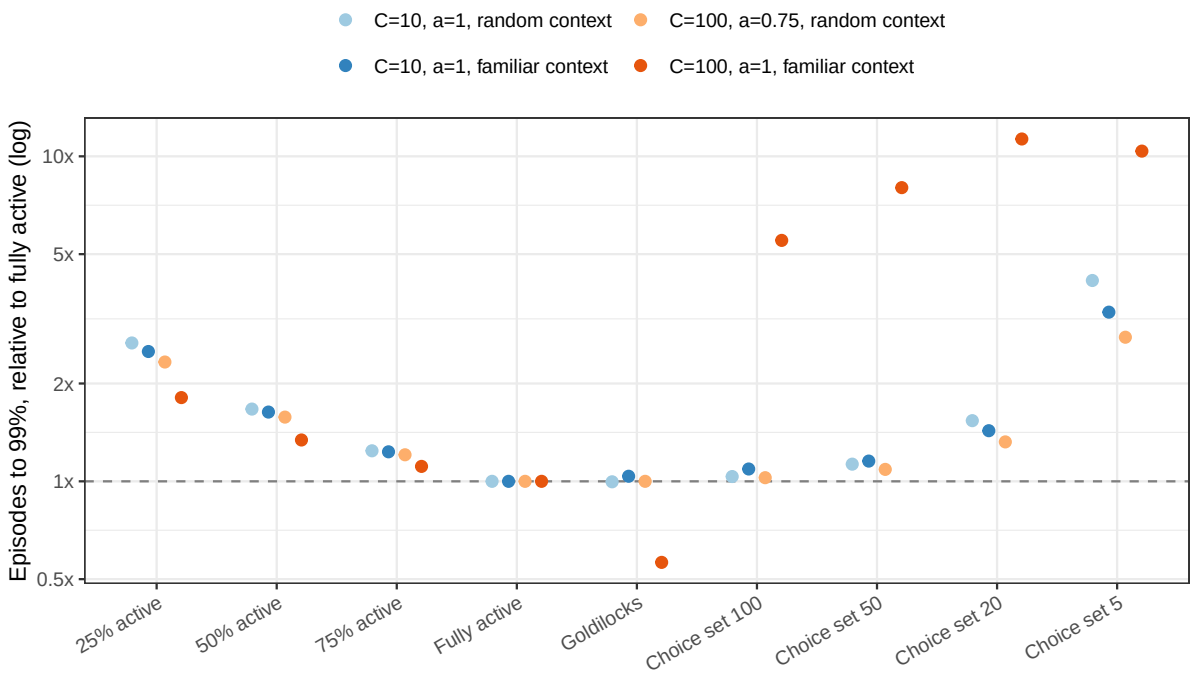


Figure 4. Episodes to learn 99% of the vocabulary relative to fully active selection (log scale; dashed line = fully active), eliminative learner, for partial autonomy, Goldilocks targeting, and bounded choice sets, in four settings. Pure passive sampling (10.6–26.6x) is omitted for scale.

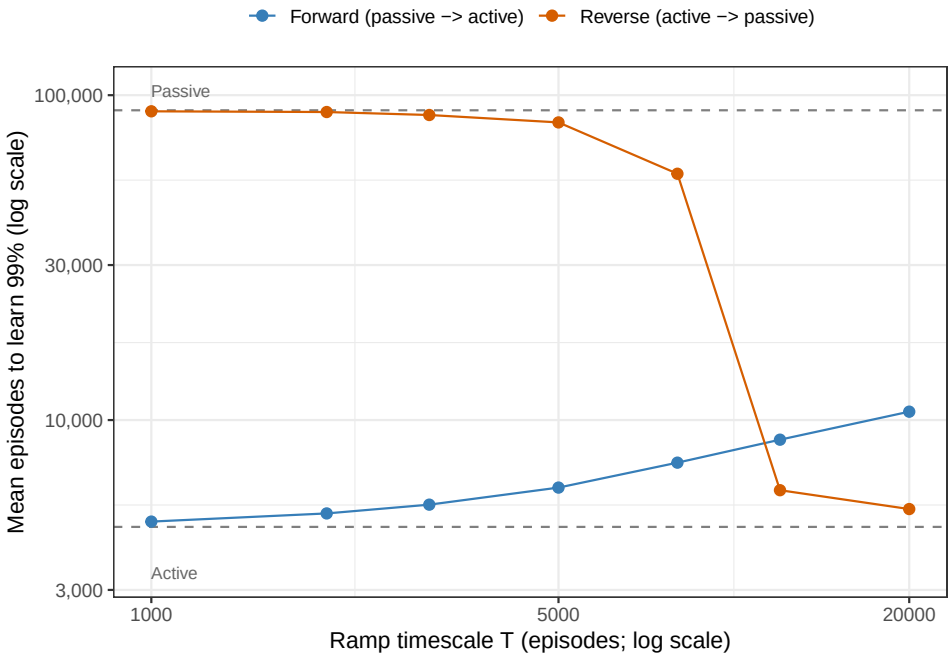


Figure 5. Mean episodes to learn 99% of the vocabulary vs. ramp timescale T, forward (passive-to-active) vs. reverse (active-to-passive) developmental trajectories. Dashed lines mark pure passive and pure active selection (Table 7).

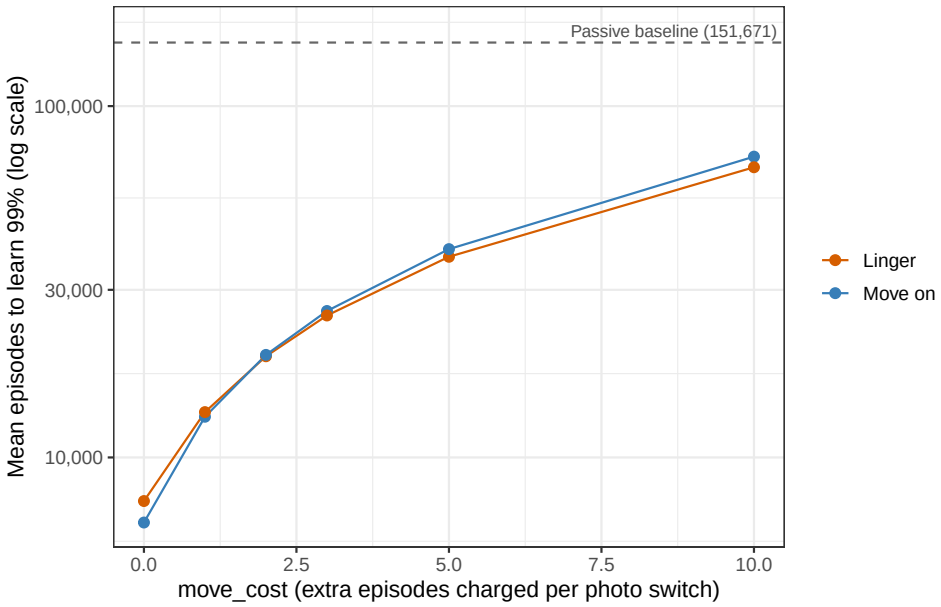


Figure 6. Mean episodes to learn 99% of the vocabulary vs. `move_cost`, active eliminative learner on Flickr8k captions. Moving on to a new photo wins when switching is cheap; lingering on the current photo overtakes it once switching costs more, with the crossover between `move_cost`=1 and 2.